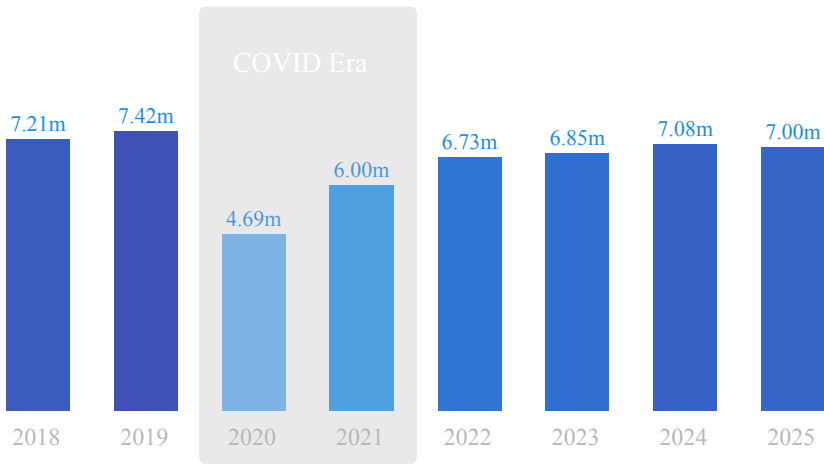


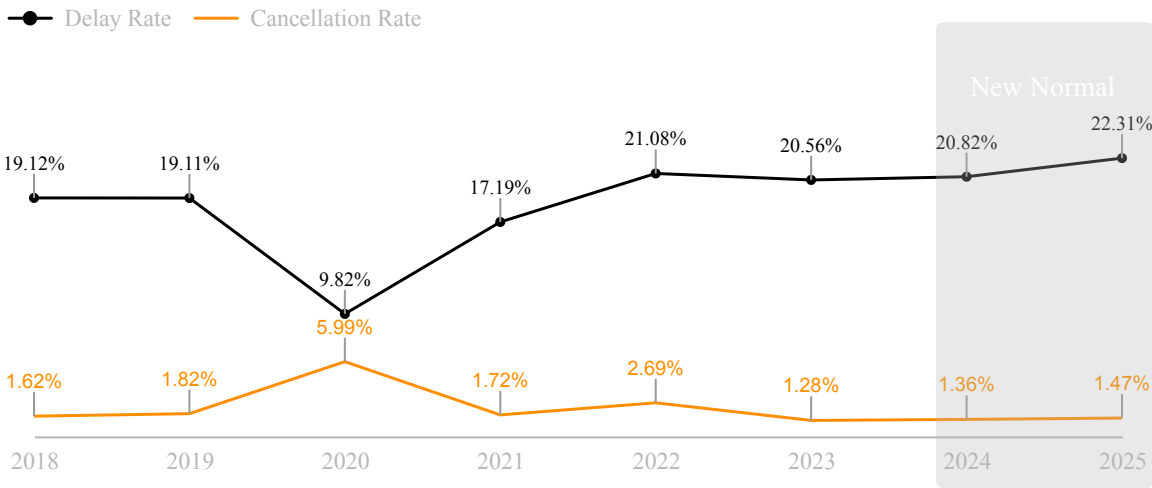
Disruptions Development, 2018–2025 – BTS-Reporting US Carriers

Years are grouped into four periods that reflect how disruption dynamics evolved: **Pre-COVID** (2018–2019), **COVID** (2020–2021), **Recovery** (2022–2023), and **New Normal** (2024–2025). On the left chart below you can see that scheduled **flight volume remains below 2019 levels** through 2024–2025, yet revenue passenger miles are ~7% above 2019 with load factor unchanged — meaning longer routes or larger aircraft, not fuller planes. (Source: BTS [Load Factor](#) and [Revenue Passenger Miles](#), via FRED). Notice how the drop in flights during COVID years also affected disruption rates — and that rates haven't fully settled in the new normal: delay rates in particular continue to climb through 2025. Charts at the bottom show **pp change** vs. the 2018–2019 baseline **for rates**, and **relative % change for volume** — pp avoids exaggerating small movements at these low rates (1–4%). 2020 shows the expected pattern: flight volume dropped sharply, cancellations rose under COVID restrictions — but reduced traffic cut delay rates by more than half. By the new normal, **cancellations sit below pre-COVID while delays run higher** — consistent with airlines running leaner schedules and less slack to absorb disruption. *Baseline = SUM(numerator)/SUM(denominator) across 2018–2019, not an average of yearly rates.*

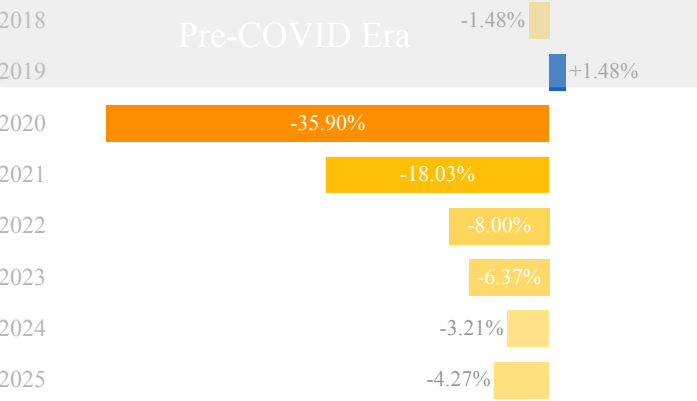
Scheduled flights development 2018 - 2025



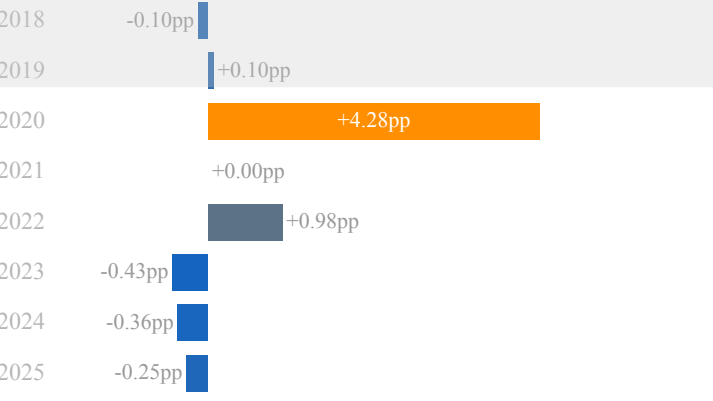
Disruptions Rate over time



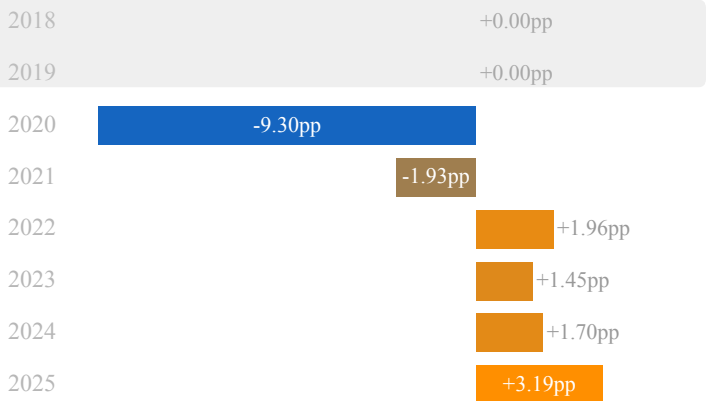
Scheduled flights % change vs Pre-COVID average



Cancellation pp Rate change vs Pre-COVID average



Delay Rate pp change vs Pre-COVID average



Pre-COVID (2018-19) vs New Normal (2024-25) Era Comparison on Carrier Level

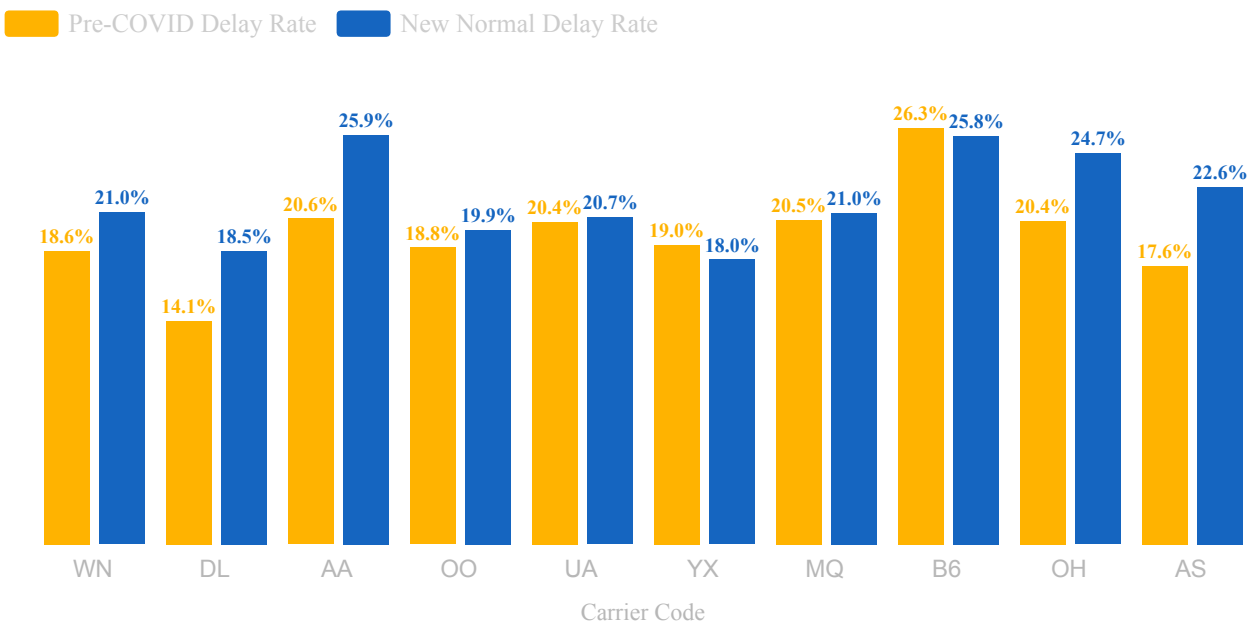
There are no major shifts in scheduled flight distribution between the **dataset's 19 carriers**, so for simplicity the table on the left shows only New Normal-era data. Worth noting: the **top 3 carriers operate nearly 50% of reported flights**, and the **top 5 account for over 70%** — worth keeping in mind when reading the carrier-level rate changes below. **Changes in delay rate are fairly evenly distributed across carriers**, mostly following the overall trend — though **American (AA), Alaska (AS), and PSA (OH)** show a stronger increase than the rest. Cancellation rate shows more varied movement: carriers ranked 6th and below often run notably higher rates than the top 5. The largest carrier, **Southwest (WN) and Envoy Air (MQ) both show significant improvement**. Delta shows the opposite pattern — **its cancellation rate has more than tripled**, yet remains well below the industry average. Rather than driving the current rate up, **Delta's low pre-COVID baseline was pulling the average down**.

	Cancellation Rate	Delay Rate
Pre-COVID	1.72%	19.12%
New Normal	1.41%	21.56%

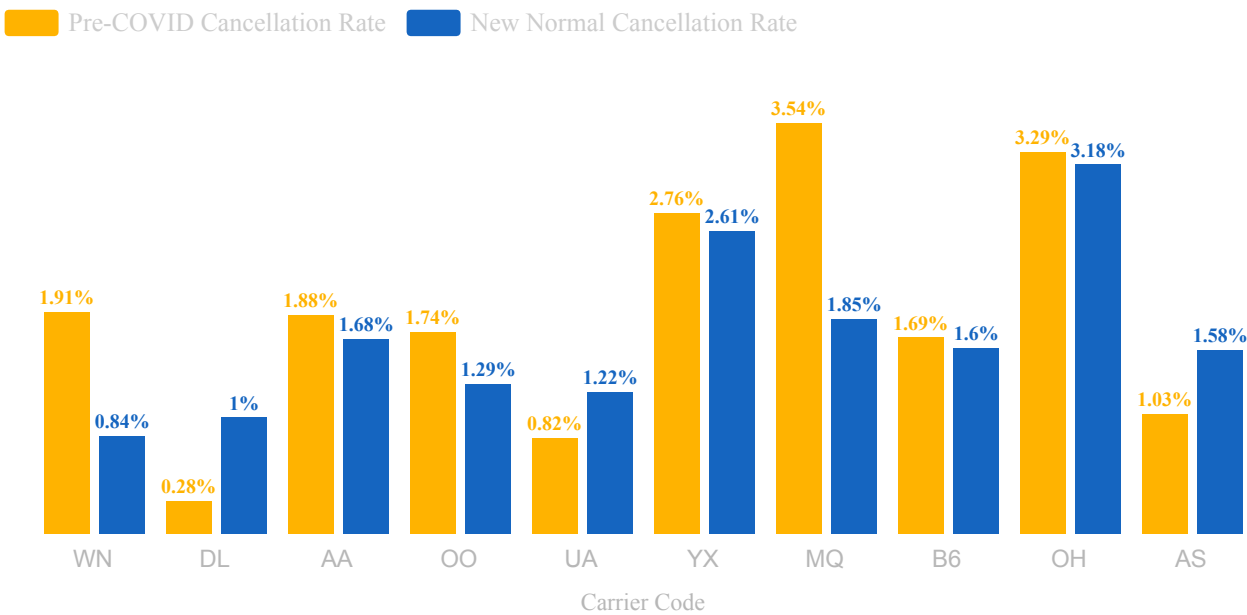
New Normal — TOP 10 Carriers Scheduled Flights

	Code	Name	Scheduled Flights ▾	% of Total
1.	WN	Southwest Airlines ...	2,811,304	19.97%
2.	DL	Delta Air Lines Inc.	2,035,506	14.46%
3.	AA	American Airlines Inc.	1,957,959	13.91%
4.	OO	SkyWest Airlines Inc.	1,584,479	11.25%
5.	UA	United Air Lines Inc.	1,555,722	11.05%
6.	YX	Republic Airline	647,501	4.6%
7.	MQ	Envoy Air	579,277	4.11%
8.	AS	Alaska Airlines Inc.	491,407	3.49%
9.	OH	PSA Airlines Inc.	476,706	3.39%
10.	B6	JetBlue Airways	471,695	3.35%
	Grand total		14,080,680	100%

Pre-COVID vs New Normal Delay Rate by Carrier



Pre-COVID vs New Normal Cancellation Rate by Carrier

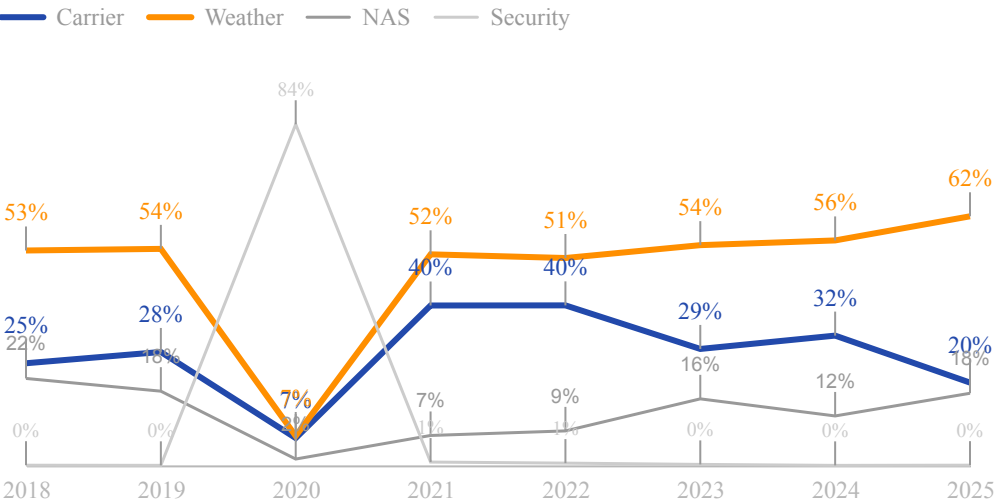


Disruption categories – development & carrier decomposition

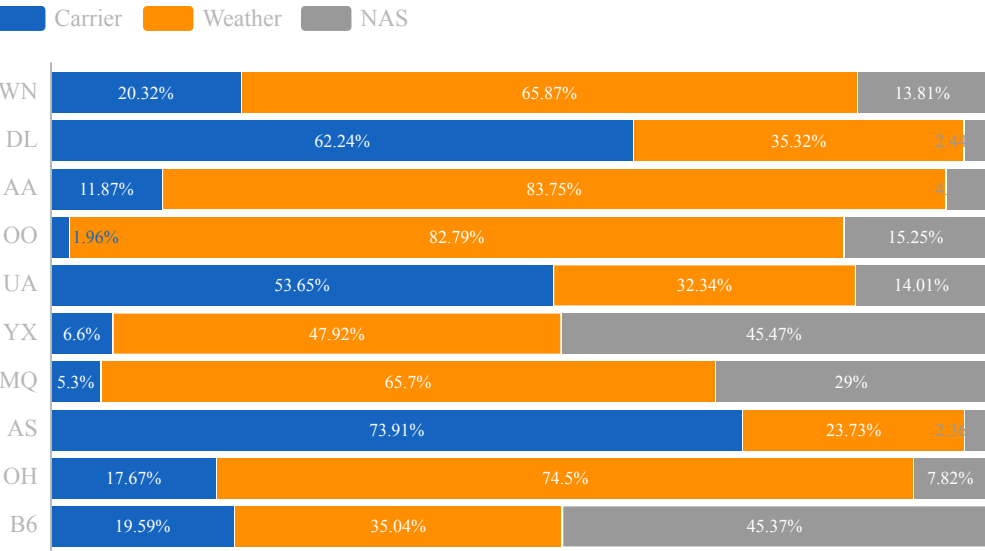
During normal years, **weather is at least 2× stronger a cancellation factor than carrier operations**. During COVID and recovery, the carrier share rose sharply — reflecting the operational difficulties carriers faced in a fast-evolving environment. **Security cancellations spiked to 84% in 2020** — at that scale this reflects government-mandated travel restrictions and border closures rather than typical TSA screening incidents, suggesting carriers classified COVID policy measures under this category. For delays the picture is different: **weather has minor impact once a flight has departed**. Late-aircraft delays dominate in normal years — not a root cause in itself, but a cascading effect of lean scheduling: carrier delays on one rotation — maintenance, late crew, fueling, and similar factors within the airline's control — propagate to the next flight via the same aircraft. Outside COVID, it is **this carrier-to-late-aircraft cascade — not weather or NAS — that drives the most delay minutes**.

Looking at carrier decomposition in the bottom charts, **Delta (DL), American (AA), and Alaska (AS) show unusually high carrier cancellation shares (over 50%)**, suggesting meaningful operational stress or schedule restructuring during the period. On the delay side the picture is more evenly distributed — the only standout shares worth noting are **carrier delays at Delta (DL) and SkyWest (OO)**.

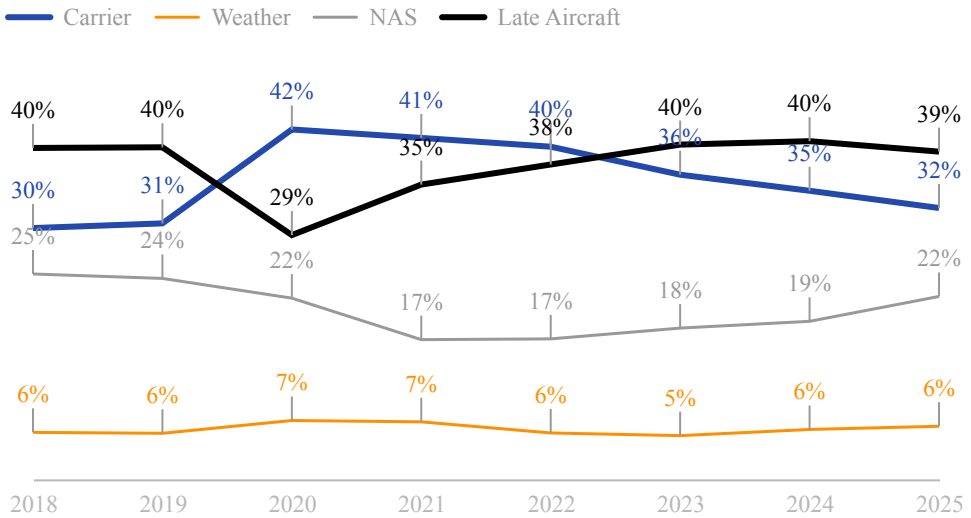
Cancellation Categories Over Time



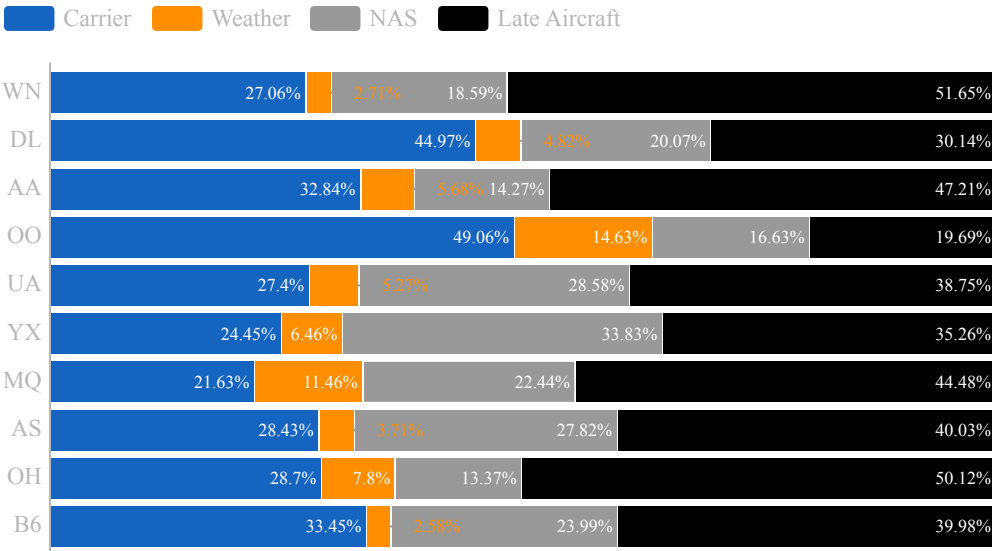
Cancellation Categories per Carrier — New Normal (2024–25)



Delay Categories Over Time



Delay Categories per Carrier — New Normal (2024–25)



Pre-COVID (2018-19) vs New Normal (2024-25) Era Comparison on Airport Level

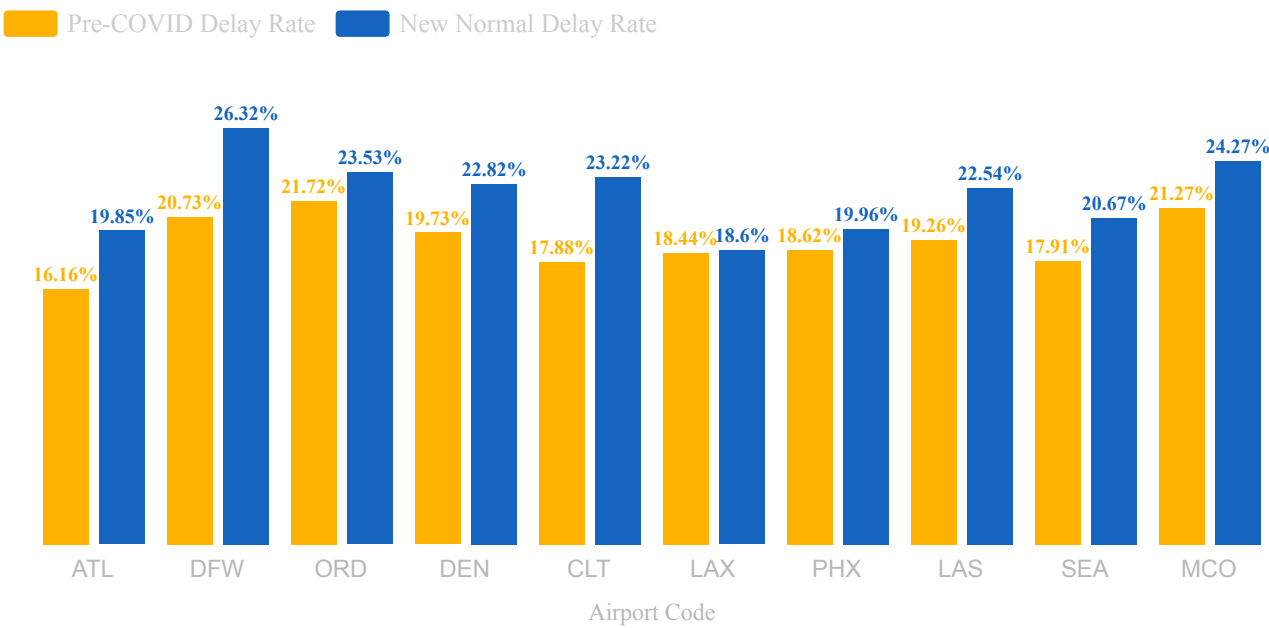
There are **390 airports** in the dataset across the full 2018–2025 range; the table shows the top 10 by New Normal volume, plus all remaining airports grouped as "Others." **The most disruptive airport has changed.** Pre-COVID, **Chicago** (ORD) and **Dallas** (DFW) had similar delay rates, with Dallas slightly better on cancellation. By the New Normal, Dallas's delay rate jumped over 5pp — now the highest of any major airport — while cancellation stayed flat. Chicago moved the other way: delay rose only modestly, and cancellation improved by more than a full percentage point. **Dallas is now the most disruptive airport** on both measures. Beyond that shift, **delay rate rose** at nearly every major airport, while **cancellation rate improved** at most — Atlanta (ATL), the largest airport by volume— is the one exception, with cancellations nearly doubling. Even so, Atlanta remains below the industry average on both measures: proof that **scale alone doesn't force worse performance**, if congestion and connections are managed well.

	Cancellation Rate	Delay Rate
Pre-COVID	1.72%	19.12%
New Normal	1.41%	21.56%

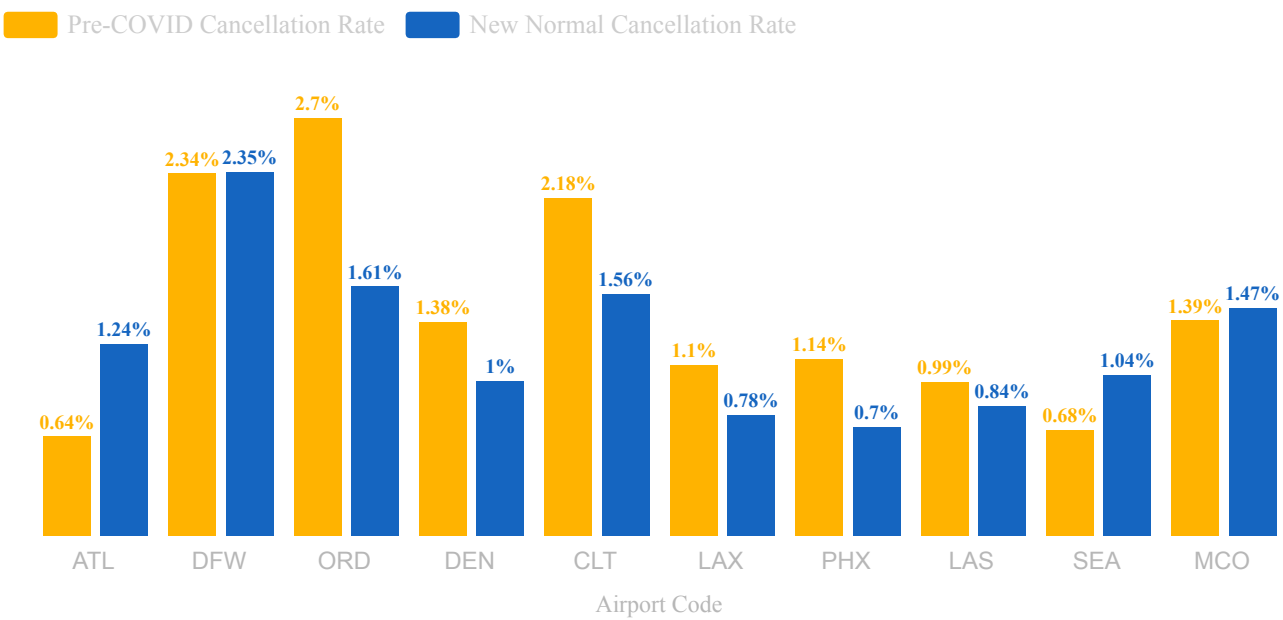
New Normal — TOP 10 Airports Scheduled Flights

	Code	City	Scheduled Flights	% of Total	Disruptions Count
1.	ATL	Atlanta, GA	5,639,473	4.66%	999,671
2.	DFW	Dallas/Fort Worth...	4,868,733	4.47%	1,175,713
3.	ORD	Chicago, IL	4,819,842	4.31%	1,088,522
4.	DEN	Denver, CO	4,581,137	4.45%	1,036,859
5.	CLT	Charlotte, NC	3,460,759	2.93%	699,325
6.	LAX	Los Angeles, CA	3,206,143	2.73%	608,395
7.	PHX	Phoenix, AZ	2,933,113	2.77%	585,271
8.	LAS	Las Vegas, NV	2,806,439	2.65%	649,385
9.	SEA	Seattle, WA	2,593,925	2.33%	480,660
10.	MCO	Orlando, FL	2,447,091	2.27%	610,034
11.	Others		75,559,465	66.42%	15,707,329
	Grand total		112,916,120	100%	23,641,164

Pre-COVID vs New Normal Delay Rate by Airport



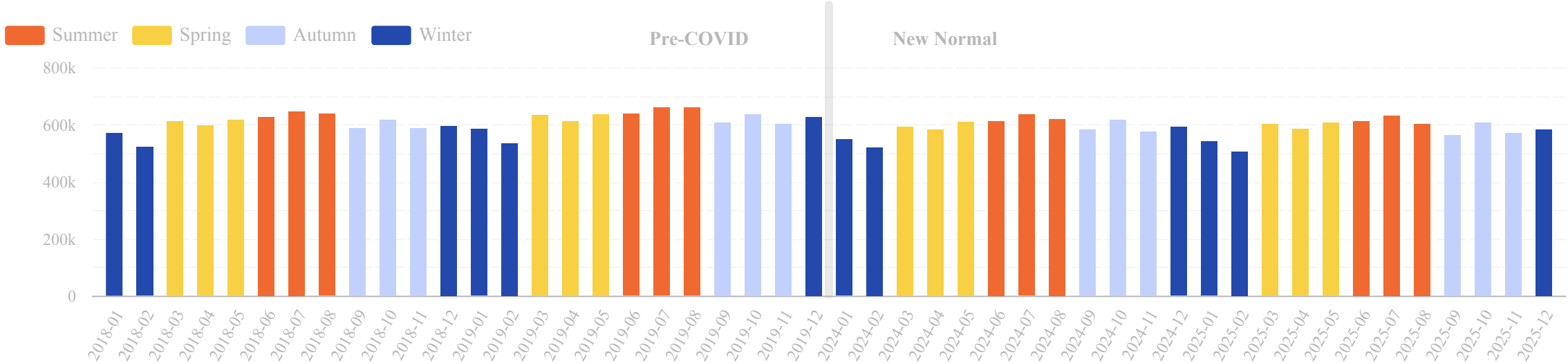
Pre-COVID vs New Normal Cancellation Rate by Airport



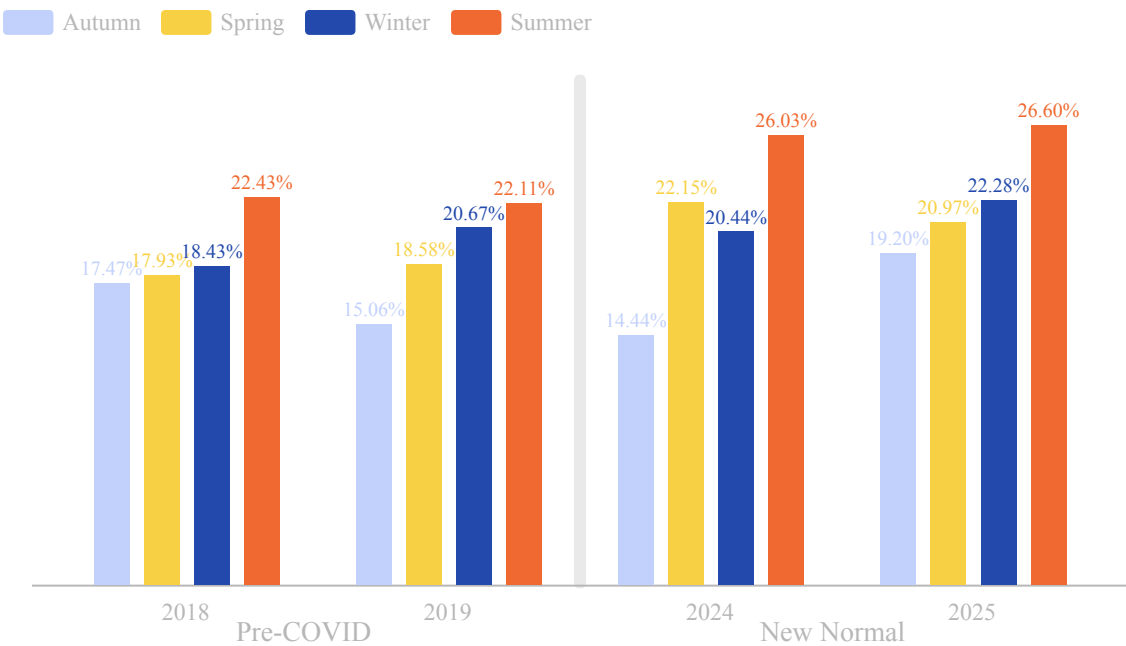
Pre-covid vs New Normal Seasonal Patterns

COVID and Recovery years (2020–2023) are excluded to avoid diluting the underlying seasonal signal. Scheduled flight volume is more evenly distributed across the year than most people expect — the gap between the busiest month (July) and the quietest (February) is **only ~20%**, reflecting the mature, route-dense nature of domestic US aviation. **Delay rates follow a clear seasonal pattern peaking in summer.** The summer peak has intensified in the new normal — **summer 2025 reached 26.6% vs 22.1%** in the last pre-COVID summer — though the underlying drivers would require weather and scheduling data to confirm. **Cancellation rates tell a different story:** winter shows elevated cancellations despite being the quietest season, consistent with weather-driven groundings that occur regardless of traffic volume. The summer cancellation spike in 2019 and 2025 stands out and would warrant further investigation with weather data to separate cause from coincidence.

Scheduled Flights Monthly



Delay Rate by Year and Season



Cancellation Rate by Year and Season

